ORTHOGONALITY · QR

Where each factor comes from

Running Gram–Schmidt on the columns of A produces both factors at once. Q is the output and R is the bookkeeping — the coefficients that were subtracted along the way, kept rather than discarded.

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FACTORS

TYPE	CONDITION	ENTRIES	PROPERTIES	WHAT IT RECORDS
THE OUTPUT				
 **Q — orthonormal columns**	$m \times n$	$\mathbf{q}_j = \mathbf{u}_j / \lVert\mathbf{u}_j\rVert$	$Q^{\mathsf{T}}Q = I_n$	an orthonormal basis for $\mathrm{Col}(A)$
THE BOOKKEEPING				
 **R — upper triangular**	$n \times n$	$r_{ij} = \mathbf{q}_i \cdot \mathbf{v}_j$	diagonal $r_{jj} = \lVert\mathbf{u}_j\rVert > 0$	the coefficients discarded during orthogonalization

This is why QR is not a separate algorithm from Gram–Schmidt so much as its record. Every projection coefficient computed during orthogonalization is an entry of R, and R is triangular precisely because $\mathbf{v}_j$ is only ever projected onto vectors that came before it.